

Postprandial glucose, insulin, and free fatty acid responses to sucrose consumed with blackcurrants and lingonberries in healthy women.

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BACKGROUND: Sucrose induces high postprandial glucose and insulin responses. In vitro studies suggest that berries may reduce the digestion and absorption of sucrose and thereby suppress postprandial glycemia, but the evidence in humans is limited. OBJECTIVE: We investigated the effects of sucrose ingested with blackcurrants (*Ribes nigrum*) and lingonberries (*Vaccinium vitis-idaea*) on postprandial glucose, insulin, and free fatty acid responses. DESIGN: Twenty healthy women participated in a randomized, controlled, crossover meal study. They consumed whole blackcurrants or lingonberries (150 g served as purées) or blackcurrant or lingonberry nectars (300 mL), each with 35 g added sucrose. Sucrose alone (35 g in 300 mL water) was used as a reference. Blood samples were collected at 0, 15, 30, 45, 60, 90, and 120 min. RESULTS: In comparison with sucrose alone, ingestion of sucrose with whole berries resulted in reduced glucose and insulin concentrations during the first 30 min and a slower decline during the second hour and a significantly improved glycemic profile. Berries prevented the sucrose-induced late postprandial hypoglycemic response and the compensatory free fatty acid rebound. Nearly similar effects were observed when sucrose was consumed with berry nectars. The improved responses were evident despite the higher content of available carbohydrate in the berry and nectar meals, because of the natural sugars present in berries. CONCLUSIONS: Blackcurrants and lingonberries, as either whole berries or nectars, optimize the postprandial metabolic responses to sucrose. The responses are consistent with delayed digestion of sucrose and consequent slower absorption of glucose.